Typical Workflow for Computing Superconducting Observables from Migdal–Eliashberg Theory

Abstract

This note summarizes a standard first-principles Migdal–Eliashberg (ME) workflow when the harmonic phonon dispersion $\omega_{\mathbf{q}\nu}$ and bare electron–phonon matrix elements $g_{mn\nu}(\mathbf{k},\mathbf{q})$ are already available. The main steps are: construct the Fermi-surface-resolved electron–phonon coupling, form the Eliashberg spectral function $\alpha^2F(\omega)$ and related moments, choose a Coulomb treatment, solve the isotropic or anisotropic ME equations on the Matsubara axis, determine T_c, and analytically continue to the real axis when spectroscopic observables are required. Common outputs include the superconducting gap, quasiparticle renormalization, tunneling density of states, specific heat, condensation energy, critical fields, isotope coefficients, and strong-coupling ratios. A dedicated section constructs an effective Bogoliubov–de Gennes Hamiltonian directly in the same Wannier basis used for electron–phonon interpolation, including matrix self-energies and real-space pairing amplitudes.

1 Inputs and conventions

Assume the normal-state electronic structure $\epsilon_{n\mathbf{k}}$, phonon frequencies $\omega_{\mathbf{q}\nu}$, and bare electron–phonon matrix elements

$$g_{mn\nu}(\mathbf{k},\mathbf{q}) = \langle m,\mathbf{k}+\mathbf{q}|\delta_{\mathbf{q}\nu}V_{\mathrm{KS}}|n,\mathbf{k}\rangle \tag{1}$$

are known. Here n,m are electronic bands, ν is a phonon branch, and $\mathbf{q}$ is the phonon wavevector. In a density-functional perturbation theory implementation the normalization of g usually already contains the phonon zero-point displacement factor $\sqrt{\hbar/(2M\omega_{\mathbf{q}\nu})}$; one should verify the code convention before inserting g into formulas.

The ME approximation assumes that the dominant pairing interaction is the retarded electron–phonon interaction and that vertex corrections are small in the Migdal sense, schematically of relative order $\omega_{\mathrm{ph}}/E_{\mathrm{F}}$ for a conventional metal. The equations are then expressed in terms of a normal self-energy and an anomalous pairing self-energy.

2 Step 1: Resolve the electron–phonon coupling on the Fermi surface

The first practical task is to convert the microscopic $g_{mn\nu}(\mathbf{k},\mathbf{q})$ into Fermi-surface scattering weights. A commonly used phonon linewidth is

$$\gamma_{\mathbf{q}\nu} = 2\pi\omega_{\mathbf{q}\nu}\sum_{nm\mathbf{k}}|g_{mn\nu}(\mathbf{k},\mathbf{q})|^2\,\delta(\epsilon_{n\mathbf{k}}-E_{\mathrm{F}})\delta(\epsilon_{m,\mathbf{k}+\mathbf{q}}-E_{\mathrm{F}}), \tag{2}$$

up to the Brillouin-zone normalization convention of the code.

The corresponding mode-resolved dimensionless coupling can be written

$$\lambda_{\mathbf{q}\nu} = \frac{\gamma_{\mathbf{q}\nu}}{\pi N(0)\omega_{\mathbf{q}\nu}^2}, \tag{3}$$

where $N(0)$ is the electronic density of states at the Fermi level, usually per spin unless a different convention is stated.

The total coupling is

$$\lambda = \sum_{\mathbf{q}\nu} w_{\mathbf{q}} \lambda_{\mathbf{q}\nu}, \tag{4}$$

with $w_{\mathbf{q}}$ the normalized $\mathbf{q}$-point weights.

Numerical issue. The double Fermi-surface delta function makes dense $\mathbf{k}$ and $\mathbf{q}$ meshes essential. In practice one normally performs Wannier interpolation of electronic bands, phonons, and g onto much denser meshes than were used in the underlying first-principles calculation. The convergence of λ and α^2F with respect to electronic smearing and mesh density should be checked explicitly.

3 Step 2: Construct the Eliashberg spectral function

The isotropic Eliashberg spectral function is

$$\alpha^2F(\omega) = \frac{1}{2}\sum_{\mathbf{q}\nu} w_{\mathbf{q}} \lambda_{\mathbf{q}\nu} \omega_{\mathbf{q}\nu} \delta(\omega - \omega_{\mathbf{q}\nu}). \tag{5}$$

Equivalently,

$$\lambda = 2\int_0^\infty d\omega\, \frac{\alpha^2F(\omega)}{\omega}. \tag{6}$$

Several moments of α^2F are useful:

$$\omega_{\log} = \exp\left[\frac{2}{\lambda}\int_0^\infty d\omega\, \frac{\alpha^2F(\omega)}{\omega}\ln\omega\right], \tag{7}$$

$$\omega_2 = \left[\frac{2}{\lambda}\int_0^\infty d\omega\, \alpha^2F(\omega)\,\omega\right]^{1/2}. \tag{8}$$

These quantities are useful for rough T_c estimates and for diagnosing which phonon frequency ranges dominate pairing.

4 Step 3: Decide whether the isotropic or anisotropic ME problem is required

4.1 Isotropic approximation

If the gap and electron–phonon coupling vary weakly over the Fermi surface, the microscopic momentum dependence can be averaged into $\alpha^2F(\omega)$. The problem then reduces to coupled scalar equations for the Matsubara-frequency-dependent renormalization function Z_n and gap Δ_n.

This approximation is often sufficient for conventional, nearly isotropic s-wave superconductors and is computationally inexpensive.

4.2 Anisotropic formulation

If the Fermi surface contains distinct sheets, the coupling is strongly momentum dependent, or the gap is expected to be anisotropic, retain Fermi-surface labels. The unknowns become $Z(\mathbf{k}, \mathrm{i}\omega_n)$ and $\Delta(\mathbf{k}, \mathrm{i}\omega_n)$, usually discretized on the Fermi surface or represented by dense interpolated $\mathbf{k}$ points.

A multiband approximation is intermediate between the fully isotropic and fully anisotropic limits: one forms band- or sheet-resolved kernels $\alpha^2 F_{ij}(\omega)$ and solves coupled Eliashberg equations for each sheet.

5 Step 4: Build the phonon-mediated pairing kernel

For fermionic Matsubara frequencies

$$\omega_n = (2n+1)\pi k_{\mathrm{B}} T, \tag{9}$$

the isotropic retarded pairing kernel is

$$\lambda_{n-m} = 2\int_0^\infty d\Omega \, \frac{\Omega \, \alpha^2 F(\Omega)}{\Omega^2 + (\omega_n - \omega_m)^2}. \tag{10}$$

This frequency dependence is the essential retardation physics that distinguishes Eliashberg theory from static BCS theory.

In an anisotropic calculation the corresponding kernel contains the Fermi-surface-resolved matrix elements,

$$\lambda_{\mathbf{k}\mathbf{k}'}(\mathrm{i}\omega_n - \mathrm{i}\omega_m), \tag{11}$$

with $\mathbf{k}$ and $\mathbf{k}'$ constrained to the Fermi surface and connected by a phonon momentum $\mathbf{q} = \mathbf{k}' - \mathbf{k}$.

6 Step 5: Include the Coulomb repulsion

The standard isotropic ME equations use a screened Coulomb pseudopotential μ^*, often treated as an input parameter. The Morel–Anderson form illustrates the retardation reduction of the bare screened Coulomb parameter μ:

$$\mu^* = \frac{\mu}{1 + \mu \ln(E_{\mathrm{el}}/\omega_{\mathrm{ph}})}. \tag{12}$$

Here E_{el} is an electronic energy scale and ω_{ph} a characteristic phonon scale.

A more first-principles treatment can replace scalar μ^* by a dynamically or statically screened Coulomb matrix element projected onto Fermi-surface states. The workflow below is unchanged in spirit; only the Coulomb kernel becomes momentum and possibly frequency dependent.

7 Step 6: Solve the imaginary-axis Migdal–Eliashberg equations

For an isotropic, particle-hole-symmetric system, a common form is

$$Z_n = 1 + \frac{\pi T}{\omega_n} \sum_m \lambda_{n-m} \frac{\omega_m}{\sqrt{\omega_m^2 + \Delta_m^2}}, \tag{13}$$

$$Z_n \Delta_n = \pi T \sum_m \left[\lambda_{n-m} - \mu^* \, \theta(\omega_c - |\omega_m|)\right] \frac{\Delta_m}{\sqrt{\omega_m^2 + \Delta_m^2}}. \tag{14}$$

Here ω_c is the Coulomb cutoff. Equivalent formulations use the anomalous self-energy $\phi_n = Z_n\Delta_n$ instead of Δ_n directly.

The two central functions have simple interpretations:

- Z_n: quasiparticle mass/energy renormalization due predominantly to electron–phonon coupling;
- Δ_n: frequency-dependent superconducting gap function.

At low Matsubara frequency in the normal state, one typically finds approximately

$$Z(0) \simeq 1 + \lambda, \tag{15}$$

so that the effective mass enhancement is approximately $m^*/m \simeq 1 + \lambda$ within the simple isotropic picture.

7.1 Determining T_c

There are two standard numerical strategies.

Linearized equation. Near T_c, $\Delta \to 0$, so the gap equation can be linearized. Then T_c is the temperature at which the largest eigenvalue of the pairing kernel reaches unity.

Nonlinear iteration. Alternatively, solve the full equations at a sequence of temperatures and locate the temperature at which the self-consistent gap vanishes.

The linearized method is generally more efficient and numerically clean for determining T_c.

8 Step 7: Use Allen–Dynes/McMillan only as a diagnostic or quick estimate

Before solving the full equations, it is common to estimate T_c using the Allen–Dynes modified McMillan formula,

$$T_c = \frac{f_1 f_2 \omega_{\log}}{1.20} \exp\left[-\frac{1.04(1+\lambda)}{\lambda - \mu^*(1+0.62\lambda)}\right]. \tag{16}$$

The factors f_1 and f_2 account for strong-coupling and spectral-shape corrections. This formula is useful for screening materials and catching implementation mistakes, but it should not replace a full ME solve when accurate observables or anisotropy are important.

9 Step 8: Analytically continue to the real axis for spectroscopy

Imaginary-axis solutions are numerically stable and are sufficient for T_c and many thermodynamic quantities. Spectroscopic observables, however, require real frequencies.

One may either solve mixed real/imaginary-axis Eliashberg equations directly or analytically continue $Z(\mathrm{i}\omega_n)$ and $\Delta(\mathrm{i}\omega_n)$ to

$$Z(\omega + \mathrm{i}0^+), \qquad \Delta(\omega + \mathrm{i}0^+). \tag{17}$$

Common continuation approaches include Padé approximants and direct real-axis integration. Padé continuation is convenient but can become unstable if the Matsubara data are noisy or insufficiently converged.

10 Step 9: Compute superconducting observables

Once Z and Δ are known, a broad set of observables becomes accessible.

10.1 Superconducting gap

The observable excitation gap is determined from the real-axis quasiparticle pole condition. A common practical definition is the positive solution of

$$\omega = \mathrm{Re}\,\Delta(\omega, T), \tag{18}$$

or, in weakly frequency-dependent cases, simply $\Delta(T) \approx \Delta(\omega = 0, T)$.

The strong-coupling ratio

$$\frac{2\Delta(0)}{k_{\mathrm{B}}T_c} \tag{19}$$

can be compared with the weak-coupling BCS value 3.53. Deviations quantify retardation and strong-coupling effects.

10.2 Quasiparticle density of states and tunneling

For an isotropic superconductor, the normalized quasiparticle density of states is approximately

$$\frac{N_s(\omega)}{N_n(0)} = \mathrm{Re}\left[\frac{\omega}{\sqrt{\omega^2 - \Delta^2(\omega)}}\right], \tag{20}$$

with the fully retarded self-energy used in accurate calculations. Strong electron–phonon coupling can produce phonon fine structure in the tunneling DOS, historically one of the key experimental signatures described by Eliashberg theory.

10.3 Electronic specific heat

The superconducting electronic free-energy difference $\Delta F = F_s - F_n$ can be computed directly from the Matsubara-axis solution. A frequently used form is

$$\Delta F = -2\pi T N(0) \sum_{n \geq 0} \left(\sqrt{\omega_n^2 + \Delta_n^2} - \omega_n\right) \left[Z_n^s - Z_n^n \frac{\omega_n}{\sqrt{\omega_n^2 + \Delta_n^2}}\right], \tag{21}$$

modulo spin and DOS conventions.

The superconducting electronic specific-heat difference then follows from

$$\Delta C(T) = -T \frac{d^2 \Delta F}{dT^2}. \tag{22}$$

One often reports the normalized jump

$$\frac{\Delta C(T_c)}{\gamma T_c}, \tag{23}$$

where γ is the normal-state Sommerfeld coefficient. Its weak-coupling BCS value is 1.43.

10.4 Thermodynamic critical field

The condensation free energy yields the thermodynamic critical field,

$$\frac{H_c^2(T)}{8\pi} = -\Delta F(T) \tag{24}$$

in Gaussian units, or the corresponding SI expression using the vacuum permeability μ_0.

10.5 Normal-state mass enhancement and Sommerfeld coefficient

The electron–phonon interaction enhances the quasiparticle mass and electronic heat-capacity coefficient. In the isotropic limit,

$$\frac{m^*}{m} \simeq 1 + \lambda, \qquad \gamma \simeq \gamma_{\rm band}(1 + \lambda). \tag{25}$$

These provide useful cross-checks against experiment.

10.6 Isotope coefficient

If the phonon spectrum is recomputed for different ionic masses, the isotope coefficient is

$$\alpha_{\rm iso} = -\frac{d\ln T_c}{d\ln M}. \tag{26}$$

The BCS value is 1/2 for a simple single-element harmonic system, while realistic ME calculations can deviate substantially because of multiple phonon branches, anharmonicity, and Coulomb effects.

11 Step 10: Optional electrodynamic and transport observables

With additional information such as Fermi velocities and impurity scattering, the ME solution can be used to obtain further observables, including

- penetration depth $\lambda_L(T)$ and superfluid density;
- optical conductivity $\sigma(\omega, T)$;
- microwave conductivity;
- quasiparticle scattering rates and lifetimes;
- coherence length estimates;
- upper critical field $H_{c2}(T)$, when magnetic-field and velocity effects are incorporated;
- normal-state electron–phonon resistivity using the transport spectral function $\alpha^2_{\rm tr}F(\omega)$.

These require more than the scalar isotropic α^2F because current and field responses depend on Fermi velocities and, in anisotropic materials, momentum-dependent scattering.

12 A practical computational pipeline

A typical first-principles workflow can be summarized as follows:

1. **Interpolate the inputs.** Use Wannier interpolation or an equivalent scheme to evaluate $\epsilon_{n\mathbf{k}}$, $\omega_{\mathbf{q}\nu}$, and $g_{mn\nu}(\mathbf{k},\mathbf{q})$ on dense Brillouin-zone meshes.
2. **Converge Fermi-surface sampling.** Check $N(0)$, λ, mode-resolved $\lambda_{\mathbf{q}\nu}$, and $\alpha^2F(\omega)$ versus $\mathbf{k}$ mesh, $\mathbf{q}$ mesh, and smearing width.
3. **Construct $\alpha^2F(\omega)$.** Compute λ, $\omega_{\log}$, ω_2, cumulative $\lambda(\omega)$, and branch- or atom-resolved contributions where useful.
4. **Choose Coulomb input.** Start with a physically reasonable μ^* for diagnostics, or compute a screened Coulomb kernel if a fully first-principles treatment is desired.
5. **Estimate T_c.** Evaluate Allen–Dynes/McMillan as a quick check.
6. **Solve the linearized ME equation.** Determine the converged T_c with a sufficiently large Matsubara cutoff and enough Matsubara frequencies.
7. **Solve the nonlinear equations below T_c.** Obtain $Z_n(T)$ and $\Delta_n(T)$ over the temperature range of interest.
8. **Continue to the real axis if needed.** Obtain $\Delta(\omega,T)$ and $Z(\omega,T)$ for tunneling and other spectroscopic quantities.
9. **Compute thermodynamics.** Evaluate $\Delta F(T)$, $C_s(T)$, the specific-heat jump, $H_c(T)$, and strong-coupling ratios.
10. **Upgrade to anisotropic ME if necessary.** If multiple gaps, strong Fermi-surface variation, or poor isotropic agreement appear, retain the momentum dependence of the electron–phonon and Coulomb kernels.

13 Step 11: Construct a BdG Hamiltonian directly in the electron–phonon Wannier basis

13.1 The working basis and target Hamiltonian

Suppose that the same set of N_W (spinful, if appropriate) Wannier orbitals $|\mathbf{R}a\rangle$ was used to interpolate the electron–phonon matrix elements. Define the corresponding Bloch operators $c_{\mathbf{k}a}$, with a fixed Bloch-sum convention throughout. The pre-existing normal-state Wannier Hamiltonian is

$$h^W_{ab}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} t^W_{ab}(\mathbf{R}), \qquad \xi^W(\mathbf{k}) = h^W(\mathbf{k}) - \mu I. \tag{27}$$

The desired result is a *Wannier-orbital* Nambu Hamiltonian, not a collection of independently assigned band gaps:

$$\boxed{\mathcal{H}^W_{\mathrm{BdG}}(\mathbf{k}) = \begin{pmatrix} h^W_{\mathrm{qp}}(\mathbf{k}) & \Delta^W_{\mathrm{qp}}(\mathbf{k}) \\ \Delta^{W\dagger}_{\mathrm{qp}}(\mathbf{k}) & -h^{WT}_{\mathrm{qp}}(-\mathbf{k}) \end{pmatrix}, \quad \Delta^W_{\mathrm{qp}}(\mathbf{k}) = -\Delta^{WT}_{\mathrm{qp}}(-\mathbf{k}).} \tag{28}$$

Here $h^W_{\rm qp}$ *includes* the chemical potential, so no additional $-\mu$ should be inserted in Eq. (28). The indices a, b may contain both orbital and spin labels; for singlet pairing the spin structure includes $i\sigma_y$, and therefore a spin-independent constant *orbital* gap should not be confused with a spinless diagonal gap.

13.2 Retain the interaction vertex in the Wannier gauge

Let $g^W_{ab,\nu}(\mathbf{k},\mathbf{q})$ denote the already interpolated phonon vertex appearing in

$$H_{e\text{-}ph} = \sum_{\mathbf{k}\mathbf{q}\nu} c^\dagger_{\mathbf{k}+\mathbf{q},a}\, g^W_{ab,\nu}(\mathbf{k},\mathbf{q})\, c_{\mathbf{k},b}(b_{\mathbf{q}\nu} + b^\dagger_{-\mathbf{q}\nu}). \tag{29}$$

The vertex must use the *same* orbital ordering, Bloch phases, centers, and spin convention as h^W. If the stored vertex is in the electronic band gauge, with $c^W_{\mathbf{k}} = U(\mathbf{k})c^{\rm band}_{\mathbf{k}}$, convert it by

$$g^W_\nu(\mathbf{k},\mathbf{q}) = U(\mathbf{k}+\mathbf{q})\, g^{\rm band}_\nu(\mathbf{k},\mathbf{q})\, U^\dagger(\mathbf{k}), \tag{30}$$

where U is square/unitary only for a matching isolated subspace; for a disentangled subspace use the actual semiunitary projector and track the associated approximation. Do not use an arbitrary independent band gauge at $\mathbf{k}$ and $-\mathbf{k}$ to define a pairing matrix.

For example, integrating out harmonic phonons gives the Matsubara propagator

$$D_\nu(\mathbf{q}, i\Omega_l) = \frac{2\omega_{\mathbf{q}\nu}}{\Omega_l^2 + \omega^2_{\mathbf{q}\nu}} \tag{31}$$

with the overall attractive sign placed in the effective interaction. Schematically, the *direct Cooper-channel* contribution for scattering $(\mathbf{k}, -\mathbf{k}) \to (\mathbf{k}+\mathbf{q}, -\mathbf{k}+\mathbf{q})$ is

$$V^{\rm ph}_{ab;cd}(\mathbf{k}+\mathbf{q}, \mathbf{k}; i\Omega_l) = -\sum_\nu D_\nu(\mathbf{q}, i\Omega_l)\, g^W_{ac,\nu}(\mathbf{k},\mathbf{q})\, g^W_{bd,\nu}(-\mathbf{k}, -\mathbf{q}), \tag{32}$$

with implicit electron-index conventions. A full implementation must antisymmetrize the interaction for indistinguishable electrons, take care of spin and time-reversal relations, and avoid double counting between direct and exchange contributions. Equation (32) is a working schematic, not a substitute for checking the vertex conventions against the implemented Nambu self-energy diagrams. Add the chosen screened Coulomb pairing kernel *in the same basis*.

13.3 Solve matrix-valued Eliashberg equations, not only scalar α^2F

Introduce the Wannier-basis Nambu spinor $\Psi_{\mathbf{k}} = (c_{\mathbf{k},1}, \ldots, c_{\mathbf{k},N_W}, c^\dagger_{-\mathbf{k},1}, \ldots, c^\dagger_{-\mathbf{k},N_W})^T$. The fully matrix-valued Nambu Dyson equation is

$$\mathcal{G}^{-1}_W(\mathbf{k}, i\omega_n) = i\omega_n I_{2N_W} - \begin{pmatrix} \xi^W(\mathbf{k}) & 0 \\ 0 & -\xi^{WT}(-\mathbf{k}) \end{pmatrix} - \begin{pmatrix} \Sigma^W_N(\mathbf{k}, i\omega_n) & \Phi^W(\mathbf{k}, i\omega_n) \\ \bar\Phi^W(\mathbf{k}, i\omega_n) & \Sigma^W_h(\mathbf{k}, i\omega_n) \end{pmatrix}. \tag{33}$$

The lower blocks are constrained by Nambu particle–hole symmetry; they are not independent adjustable inputs. Self-consistency is obtained by contracting g^W on both sides of $\mathcal{G}_W$, with the phonon kernel and Coulomb contribution. In abbreviated matrix-index notation,

$$\Sigma^{W,\rm ph}_{N,ab}(\mathbf{k}, i\omega_n) = -\frac{T}{N_q}\sum_{\mathbf{q}m\nu} D_\nu(\mathbf{q}, i\omega_n - i\omega_m)\big[g^W_\nu(\mathbf{k}-\mathbf{q},\mathbf{q}) G^W(\mathbf{k}-\mathbf{q}, i\omega_m) g^{W\dagger}_\nu(\mathbf{k}-\mathbf{q},\mathbf{q})\big]_{ab}, \tag{34}$$

where the overall sign depends on the propagator and self-energy convention. The anomalous counterpart contracts g^W with the anomalous Green's function F^W and the appropriate time-reversed/transposed vertex; its consistent channel construction is encoded by Eq. (32). Iterate Σ_N^W, Φ^W and $\mathcal{G}_W$ below T_c. This retains interorbital, interband and momentum-dependent pairing, which a scalar isotropic α^2F cannot reconstruct. In many practical codes only the *band-diagonal/Fermi-surface* approximation is solved; the next subsection explains the restriction.

13.4 Low-energy coherent reduction to static Wannier BdG parameters

After solving and, if needed, continuing to real frequency, expand

$$\mathcal{G}_W^{-1}(\mathbf{k},\omega) \simeq \omega\, \mathcal{A}_W(\mathbf{k}) - \mathcal{H}_{0,W}^{\mathrm{Nambu}}(\mathbf{k}) - \mathrm{Re}\,{\pm}_W^R(\mathbf{k},0), \quad \mathcal{A}_W = I - \partial_\omega\, \mathrm{Re}\,{\pm}_W^R\big|_{\omega=0}. \tag{35}$$

If $\mathcal{A}_W$ is Hermitian positive definite and damping is weak, set $S_W = \mathcal{A}_W^{-1/2}$. The static coherent model is

$$\boxed{\mathcal{H}_{\mathrm{BdG,qp}}^W(\mathbf{k}) = S_W(\mathbf{k}) \left[\mathcal{H}_{0,W}^{\mathrm{Nambu}}(\mathbf{k}) + \mathrm{Re}\,{\pm}_W^R(\mathbf{k},0)\right] S_W(\mathbf{k}).} \tag{36}$$

Read its upper-left and upper-right $N_W \times N_W$ blocks as h_{qp}^W and Δ_{qp}^W, respectively. Symmetrize small numerical violations of Hermiticity and particle–hole symmetry. In a simplified situation with normal-sector residue $R_e(\mathbf{k})$ and its hole partner $R_h(-\mathbf{k})$, the pairing block becomes

$$\Delta_{\mathrm{qp}}^W(\mathbf{k}) = R_e^{1/2}(\mathbf{k})\, \Phi^W(\mathbf{k},0)\, R_h^{1/2}(-\mathbf{k}), \tag{37}$$

when the frequency derivative of the anomalous self-energy can be neglected. For scalar Z, this reduces to $\Delta_{\mathrm{qp}} = \Phi/Z$, while the normal block must be renormalized consistently. An energy-dependent, strongly damped or nonanalytic self-energy cannot generally be represented by a unique static Hermitian BdG Hamiltonian.

13.5 Fourier transform to real-space hopping and pairing

With the convention $h^W(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} t^W(\mathbf{R})$,

$$t_{ab}^{\mathrm{qp}}(\mathbf{R}) = \frac{1}{N_k} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} h_{\mathrm{qp},ab}^W(\mathbf{k}), \tag{38}$$

$$\Delta_{ab}^{\mathrm{qp}}(\mathbf{R}) = \frac{1}{N_k} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \Delta_{\mathrm{qp},ab}^W(\mathbf{k}). \tag{39}$$

These arrays can be inserted directly into a real-space Wannier BdG solver with the original orbital ordering and Wannier centers. Check $t_{ab}(\mathbf{R}) = t_{ba}^*(-\mathbf{R})$ and $\Delta_{ab}(\mathbf{R}) = -\Delta_{ba}(-\mathbf{R})$, with any additional site/basis offsets handled consistently.

13.6 If the available Eliashberg solver works only in the band basis

A useful *approximation* is to reconstruct the Wannier pairing matrix from a band-space anomalous self-energy with phase information. With $c_{\mathbf{k}}^W = U(\mathbf{k}) c_{\mathbf{k}}^{\mathrm{band}}$,

$$\boxed{\Phi^W(\mathbf{k}, i\omega_n) = U(\mathbf{k})\, \Phi^{\mathrm{band}}(\mathbf{k}, i\omega_n)\, U^T(-\mathbf{k}).} \tag{40}$$

The *transpose at* $-\mathbf{k}$, rather than a Hermitian conjugate, is essential. For a purely intraband singlet model the Φ^{band} entries require a compatible time-reversal/spin convention; simply using a scalar

diagonal matrix for arbitrary spinful bands is not generally valid. Band-diagonal ME does not determine neglected interband anomalous self-energies, and an isotropic α^2F does not determine orbital pairing texture or the relative phases of multiple gaps. Apply the same change of basis to the *normal* self-energy before the low-energy reduction, and do not independently choose phases for $U(\mathbf{k})$ and $U(-\mathbf{k})$.

13.7 Concrete output and validation checklist

Save $h^W_{ab}(\mathbf{k})$, $g^W_{ab,\nu}(\mathbf{k},\mathbf{q})$ (or a reconstructible representation), the full Φ^W, Σ^W_N on the frequency mesh, μ, the Coulomb prescription, and the low-frequency residue/renormalization matrix. Export $t^{\mathrm{qp}}_{ab}(\mathbf{R})$ and $\Delta^{\mathrm{qp}}_{ab}(\mathbf{R},T)$ in *the original Wannier index ordering*. Confirm (i) gauge covariance under orbital rotations, (ii) antisymmetry and particle–hole symmetry, (iii) sensible decay of real-space pairing versus $\mathbf{R}$, and (iv) that the BdG poles near the Fermi energy agree with the coherent low-energy peaks of the full ME spectral function. The starting electron–phonon vertices and phonon frequencies also require h^W, a chemical potential and a Coulomb prescription before they determine the superconducting state.

14 Step 12: Self-consistent BdG in the same Wannier basis

14.1 Distinguish a fitted pairing interaction from an ME gap

The coherent static pairing block Δ^W_{qp} obtained in the previous section is already the output of an ME self-consistency calculation. Merely rediagonalizing a static BdG Hamiltonian with this fixed pairing block is *not* a new gap self-consistency calculation. For spatially varying superconductivity (a surface, junction, interface, or defect), introduce a *specified effective interaction in the low-energy Wannier model* and update the pairing field from the anomalous expectation values. The effective interaction is not uniquely recoverable from one bulk gap matrix; its orbital channels, electronic window, cutoff convention, and normal-state quasiparticle renormalization must be stipulated.

14.2 Real-space Wannier BdG problem

Let $I = (\mathbf{R}, a)$ combine the lattice site and Wannier-orbital labels. For a spin-rotation-invariant singlet system, choose the Nambu spinor $\Psi = (c_{1\uparrow}, \ldots, c_{N\uparrow}, c^\dagger_{1\downarrow}, \ldots, c^\dagger_{N\downarrow})^T$. The effective Hamiltonian in this fixed Wannier gauge is

$$\mathcal{H}_{\mathrm{BdG}} = \begin{pmatrix} h^W_{\mathrm{qp}} - \mu I & \Delta^W \\ \Delta^{W\dagger} & -\big(h^W_{\mathrm{qp}} - \mu I\big)^T \end{pmatrix}, \qquad \mathcal{H}_{\mathrm{BdG}} \begin{pmatrix} u_n \\ v_n \end{pmatrix} = E_n \begin{pmatrix} u_n \\ v_n \end{pmatrix}. \tag{41}$$

Here h^W_{qp} may be replaced by h^W_0 only if the effective pairing interaction is consistently calibrated for that choice. For translational invariance, Fourier transform the orbital blocks and replace the lower-right entry with $-h^{WT}_{\mathrm{qp}}(-\mathbf{k}) + \mu I$. For spin–orbit coupling, magnetism, or triplet pairing, use the complete spin-orbital Nambu basis and the corresponding fermionic antisymmetry $\Delta(\mathbf{k}) = -\Delta^T(-\mathbf{k})$; the abbreviated singlet block above then no longer suffices.

14.3 Anomalous density and mean-field update

Define a normalized set of singlet pair operators B_P (including onsite, interorbital, and intercell channels), and a Hermitian pairing-channel interaction matrix V_{PQ} through

$$H_{\rm int} = -\sum_{PQ} V_{PQ} B_P^\dagger B_Q. \tag{42}$$

The mean-field gap satisfies

$$\boxed{\Delta_P = \sum_Q V_{PQ} F_Q, \qquad F_Q = \langle B_Q \rangle.} \tag{43}$$

For $B_{IJ} = c_{J\downarrow} c_{I\uparrow}$, the anomalous amplitude in our spin-reduced convention can be constructed from the positive BdG energies as

$$F_{IJ} = \sum_{E_n>0} u_n(I) v_n^*(J) \tanh\left(\frac{E_n}{2k_{\rm B}T}\right), \tag{44}$$

provided the eigenstate set and pairing indices are counted consistently. Project F_{IJ} onto the selected normalized singlet channels, enforcing $F_{IJ} = F_{JI}$ in the orbital–spatial part. Equivalently evaluate the anomalous Nambu Green function in the chosen channel convention:

$$F_P = T\sum_n \left[\mathcal{G}_{12}(\mathrm{i}\omega_n)\right]_P, \qquad \mathcal{G}(z) = [z - \mathcal{H}_{\rm BdG}]^{-1}, \tag{45}$$

where the overall sign and any factors of $k_{\rm B}$ follow the Green-function and pair-operator convention (the equation uses energy units $k_{\rm B} = 1$). The Green-function approach can avoid full diagonalization when only a small number of matrix elements is required.

For iteration m, form $F^{(m)}$, compute $\Delta_P^{\rm target} = \sum_Q V_{PQ} F_Q^{(m)}$, and mix

$$\Delta_P^{(m+1)} = (1-\eta)\Delta_P^{(m)} + \eta \Delta_P^{\rm target}, \qquad 0 < \eta \le 1. \tag{46}$$

Recompute the chemical potential if the electron number is fixed. Check convergence in both the pairing matrix and particle number, and compare solutions from multiple initial gaps to detect metastable states. If charge/Hartree terms are also iterated, remove any normal-state contributions already present in the input h^W to avoid double counting.

14.4 Obtaining V_{PQ} from the Eliashberg calculation

A convenient practical choice is to fit a *static, low-energy* kernel $V_{PQ}^{\rm eff}$ in a limited set of Wannier pair channels. The exact phonon-mediated Cooper interaction generated by the interpolated vertices is retarded: schematically

$$\mathcal{V}_{PQ}^{\rm ph}(\mathrm{i}\Omega) \sim -\sum_{\mathbf{q}\nu} \mathcal{G}_{P,\mathbf{q}\nu} \mathcal{G}_{Q,\mathbf{q}\nu}^* \frac{2\omega_{\mathbf{q}\nu}}{\Omega^2 + \omega_{\mathbf{q}\nu}^2}, \tag{47}$$

with pair-channel form factors $\mathcal{G}$ built from the *same* Wannier-gauge electron–phonon vertices; phonon energies are used as frequency variables. The exact momentum and orbital arrangement must follow the scattering and time-reversal conventions of the code. Replacing this dynamic interaction by $V^{\rm eff}$ requires a specified energy window or cutoff, and should be regarded as a model reduction, not an identity.

Fit V^{eff} so that the homogeneous *self-consistent* BdG solution reproduces as much ME information as possible: T_c, the low-energy gap on each Fermi surface, its orbital and momentum structure, and ideally $\Delta(T)$. A fit solely to one value of $\Delta(0)$ is underdetermined. For an onsite diagonal approximation the update is $\Delta_{Ia,Ia} = U_a^{\mathrm{eff}} F_{Ia,Ia}$, but general multiorbital materials may need nonlocal and interorbital pair channels. The kernel must be refitted if one switches from h_0^W to h_{qp}^W or changes its electronic cutoff.

An alternative is to retain the full dynamical anomalous self-energy:

$$\Phi_{ab}^W(\mathbf{k}, \mathrm{i}\omega_n) = -k_{\mathrm{B}}T \sum_{\mathbf{k}',m,cd} \mathcal{V}_{ab,cd}(\mathbf{k}, \mathbf{k}'; \mathrm{i}\omega_n - \mathrm{i}\omega_m) F_{cd}^W(\mathbf{k}', \mathrm{i}\omega_m). \tag{48}$$

Together with the normal self-energy update, this is a matrix-valued Eliashberg/Nambu Green-function calculation, *not* ordinary static BdG self-consistency.

14.5 Recommended implementation and validation

1. Choose h_0^W or the ME-renormalized h_{qp}^W and fix its Wannier orbital ordering and gauge.
2. Select onsite/interorbital/intersite singlet pair channels and an energy window; fit V^{eff} from homogeneous ME results.
3. Solve homogeneous BdG self-consistently and validate T_c, the Fermi-surface gap, and temperature dependence against ME.
4. Build the real-space Wannier Hamiltonian for a slab, interface, defect, or Josephson geometry; initialize the pairing field from the bulk solution in superconducting regions.
5. Iterate Eqs. (43)–(44) with mixing, number conservation and appropriate boundary/lead conditions.
6. From converged solutions compute Δ_{IJ}, the anomalous density, local tunneling density of states, proximity profiles, Andreev states, and (when phases/boundaries are treated consistently) the Josephson current–phase relation.

This procedure is a physically calibrated, static low-energy approximation to ME; it does not preserve its full frequency dependence.

15 Constructing the static BdG interaction from electron–phonon vertices

The Wannier-gauge coupling $g_{ab}^{\nu}(\mathbf{k}, \mathbf{q})$ is the microscopic input, but the *pairing potential* Δ is not itself a two-body interaction. Integrating out harmonic phonons produces a retarded four-fermion vertex. With phonon energies used for frequency variables and the convention $D_{\nu}^0(\mathbf{q}, \mathrm{i}\Omega) = -2\omega_{\mathbf{q}\nu}/(\Omega^2 + \omega_{\mathbf{q}\nu}^2)$, the schematic single-band singlet Cooper kernel is

$$V^{\mathrm{ph}}(\mathbf{k}, \mathbf{k}'; \mathrm{i}\Omega) = -\sum_{\nu} \frac{2|g^{\nu}(\mathbf{k}, \mathbf{k}' - \mathbf{k})|^2 \omega_{\mathbf{k}'-\mathbf{k},\nu}}{\Omega^2 + \omega_{\mathbf{k}'-\mathbf{k},\nu}^2}. \tag{49}$$

The general Wannier-orbital kernel has four indices $V_{ab;cd}^{\mathrm{ph}}(\mathbf{k}, \mathbf{k}'; \mathrm{i}\Omega)$: it is formed from *two* vertices acting on the time-reversed electrons of a Cooper pair. Correct ordering of indices, conjugation,

fermionic antisymmetrization, and spin projection is required; a naive elementwise $|g_{ab}|^2$ does not define the multiorbital Cooper kernel.

The simplest instantaneous approximation is the static limit

$$V_{\rm stat}^{\rm ph}(\mathbf{k},\mathbf{k}') = -\sum_{\nu} \frac{2|g^{\nu}(\mathbf{k},\mathbf{k}'-\mathbf{k})|^2}{\omega_{\mathbf{k}'-\mathbf{k},\nu}}, \tag{50}$$

for the single-band case. Fourier transformation and projection onto normalized Wannier-pair channels B_P lead to a model $H_{\rm pair} = \sum_{PQ} V_{PQ}^{\rm eff} B_P^\dagger B_Q$, and the signed-interaction gap equation

$$\Delta_P = -\sum_{Q} V_{PQ}^{\rm eff} \langle B_Q \rangle. \tag{51}$$

Here $V^{\rm eff} < 0$ corresponds to attraction; if instead $U = -V^{\rm eff} > 0$ is called the attractive coupling, the same equation reads $\Delta_P = \sum_Q U_{PQ} F_Q$. A static limit by itself is *not* an exact reduction of ME: it loses retardation, normal self-energy renormalization, and frequency-dependent Coulomb screening. The pairing cutoff or active electronic space must be specified. The Coulomb repulsion (often represented in ME by μ^*) must be included without double counting.

A practical controlled approximation is to project the microscopic kernel onto dominant local and bond pairing channels, fit a low-energy static $V^{\rm eff}$ (with fixed cutoff and either bare-electron or quasiparticle normal Hamiltonian) to homogeneous ME results such as T_c, gap magnitude, and gap anisotropy, and then hold $V^{\rm eff}$ fixed while solving inhomogeneous real-space BdG self-consistently. A gap-only fit does not uniquely determine the interaction. Keeping the retarded kernel instead requires a real-space, frequency-dependent Nambu–Eliashberg Green-function calculation rather than ordinary static BdG.

16 Local characterization of the converged Wannier BdG state

Let $I = (i, a)$ and $J = (j, b)$ jointly label cells and Wannier orbitals. Three complementary outputs are the pairing potential Δ_{IJ}, the anomalous density $F_{IJ} = \langle c_{J\downarrow} c_{I\uparrow} \rangle$, and the local quasiparticle density of states. In an induced normal region, $V^{\rm eff} = 0$ can imply $\Delta = 0$ even when $F \neq 0$; thus F is the appropriate proximity-pair-correlation diagnostic.

For a complete set of positive BdG energies in a spin-singlet, spin-degenerate convention, the onsite-resolved spectral function is

$$\rho_I(E) = \sum_{n:E_n>0} \big[|u_n(I)|^2 \delta(E - E_n) + |v_n(I)|^2 \delta(E + E_n) \big], \tag{52}$$

with a spin sum if the total rather than single-spin DOS is desired. For visualizations broaden the delta functions, explicitly reporting the broadening. A spectroscopic local gap may be estimated from the low-energy onset of $\rho_I(E)$; unlike $|\Delta_I|$, it reflects Andreev bound states and other in-gap excitations. A threshold-based gap estimate depends on broadening and threshold.

The same localized Wannier functions used to interpolate the electronic structure reconstruct the continuum anomalous amplitude:

$$F(\mathbf{r},\mathbf{r}') = \sum_{IJ} w_I(\mathbf{r}) w_J(\mathbf{r}') F_{IJ}, \qquad F(\mathbf{r},\mathbf{r}) = \sum_{IJ} w_I(\mathbf{r}) w_J(\mathbf{r}) F_{IJ}. \tag{53}$$

The equal-position amplitude is useful for onsite singlet correlations, but need not characterize non-s-wave pairing. In a fixed Wannier gauge one can also average $|F_{IJ}|$ over orbitals and bonds in a

given layer, and compare $|\Delta_{IJ}|$, local DOS, and pair amplitudes across the same interface. Individual orbital-resolved entries depend on Wannier gauge; continuum quantities and appropriately defined physical responses are preferable for direct experimental comparisons.

For a normal-metal proximity profile, a fit $|F(z)| \propto \exp(-z/\xi_N)$ estimates a decay length only when the data support exponential decay; clean-limit or multichannel profiles can be oscillatory or nonexponential. The complex phase of a dominant gap channel, $\Delta_I = |\Delta_I| e^{\mathrm{i}\varphi_I}$, describes spatial phase textures; the bond current can be computed from the normal density matrix and Peierls-dressed hoppings. With $e > 0$ denoting elementary charge, a common orientation convention gives

$$I_{I\to J} = \frac{2e}{\hbar} \operatorname{Im}\left[h_{IJ} \sum_{\sigma} \langle c^{\dagger}_{I\sigma} c_{J\sigma} \rangle \right], \tag{54}$$

where the sign depends on the stated orientation and electron-charge convention. Imposed electrode phase differences yield the Josephson $I(\phi)$ and $I_c = \max_{\phi} |I(\phi)|$. If reporting a local condensation-energy profile, specify a convention for distributing nonlocal hopping/pair energies among sites and compare superconducting and normal thermodynamic potentials with consistent mean-field constants; the total condensation energy is unambiguous even when its local decomposition is not.

17 Recommended convergence checks

A robust calculation should report at least the following convergence tests:

- dense $\mathbf{k}$ and $\mathbf{q}$ meshes for the Fermi-surface double delta function;
- electronic smearing or tetrahedron parameters;
- phonon interpolation quality and absence of spurious unstable modes;
- Matsubara cutoff $\omega_{\max}$ several times larger than the maximum phonon frequency;
- number of Matsubara frequencies at the lowest temperature studied;
- sensitivity of T_c and $\Delta(0)$ to μ^*;
- real-axis broadening/continuation parameters for spectral observables;
- isotropic versus anisotropic treatment when the Fermi surface is complex.

18 What is actually "renormalized" in the ME calculation?

It is useful to distinguish the bare first-principles inputs from the superconducting self-consistency cycle. The DFPT matrix elements g and harmonic phonons define the *interaction kernel*. The ME equations then dress the electronic propagator through the electron–phonon normal self-energy and generate an anomalous self-energy below T_c.

In the simplest standard workflow, the phonon propagator entering α^2F is not updated self-consistently after superconductivity appears. Thus the calculation is self-consistent in the electronic normal and anomalous self-energies, but not necessarily in the phonon spectrum. More elaborate theories can feed the superconducting electronic response back into the phonon self-energy if required.

19 Minimal set of outputs to save

For a reproducible calculation it is useful to save

- $\alpha^2F(\omega)$ and cumulative $\lambda(\omega)$;
- total and mode/branch-resolved λ;
- $\omega_{\log}$ and ω_2;
- chosen μ^* or Coulomb kernel;
- T_c from both Allen–Dynes and the full ME solve;
- $Z_n(T)$ and $\Delta_n(T)$;
- real-axis $Z(\omega, T)$ and $\Delta(\omega, T)$ when computed;
- $\Delta(0)$, $2\Delta(0)/(k_{\mathrm{B}}T_c)$, $\Delta C/(\gamma T_c)$, and $H_c(T)$;
- effective BdG h^{eff}, Δ^{eff}, and quasiparticle weights when a static low-energy model is desired;
- all mesh, smearing, Matsubara-cutoff, and continuation parameters.

20 Conceptual summary

Starting from $g_{mn\nu}(\mathbf{k}, \mathbf{q})$ and $\omega_{\mathbf{q}\nu}$, the essential reduction is

$$\boxed{g, \omega_{\mathbf{q}\nu} \longrightarrow \alpha^2F(\omega) \longrightarrow \lambda(\mathrm{i}\omega_n - \mathrm{i}\omega_m) \longrightarrow \{Z, \Delta\} \longrightarrow \text{superconducting observables}} \tag{55}$$

for the isotropic problem. In the anisotropic problem the same chain holds, but the momentum dependence is retained throughout rather than compressed into a single scalar $\alpha^2F(\omega)$.

Standard references

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